Following the Information Bound from a Search-Window Prior with Sequential Bayesian Experiment Design in Nitrogen-Vacancy Magnetometry

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[Written last, once the simulation and lab results are in.]

I. INTRODUCTION

Nitrogen-vacancy (N-V) centers in diamond measure magnetic fields at room temperature with nanometer spatial resolution [1, 2], and are now used in settings ranging from condensed-matter physics to biology [3–5].

The most widely used readout is optically detected magnetic resonance (ODMR): the photoluminescence of the center is recorded while a microwave drive is stepped through a search window, the range of drive frequencies in which the spin resonance can lie, and the field is read from the position of the resonance. Every point of this scan takes time, so the number of measurements needed to locate the resonance to a given precision sets both the sensitivity and the bandwidth of the magnetometer. It matters most where the scan is repeated, as when a time-varying field is tracked [6] or a field is imaged with a spectrum recorded at every pixel [7, 8].

Locating the resonance is an instance of Hamiltonian learning: the parameters of a quantum system's Hamiltonian are inferred from measurements whose settings, here the microwave drive frequency, are chosen adaptively from what has been learned so far [9, 10]. For N-V centers, Hamiltonian learning and adaptive Bayesian estimation have been demonstrated mainly in the time domain, with Rabi and Ramsey sequences in which an evolution time or a readout phase is the chosen setting [11–14]. In ODMR, sequential Bayesian experiment design (SBED), which keeps a posterior over the parameters as a sequential Monte Carlo (SMC) particle cloud [9] and chooses each drive frequency to maximize the expected information gain, reaches a given precision more than an order of magnitude faster than a sweep, the uniform scan conventionally used [15].

Non-adaptive strategies reduce the number of points instead, by measuring a subset of the window and reconstructing the spectrum from it, with a low-complexity reconstruction [16] or with compressed sensing, which matches the accuracy of a full scan with a small fraction of its points [6]. Learned measurement policies [17, 18] have recently spread with reinforcement learning (RL) across quantum metrology [19–23], including N-V magnetometry [24–26].

In ODMR, adaptive and non-adaptive strategies have been shown to need far fewer measurements than a sweep, and the gain is measured against that sweep [6, 15]; learned strategies are measured against the best established heuristics [17, 21, 26]. Information bounds are already used as benchmarks in Hamiltonian learning: the Bayesian Cramér–Rao bound certifies how close an SMC estimator comes to the best precision for the measurements it has chosen [9]. The same kind of bound, the van Trees inequality [27, 28], has been tested, for a regular prior, against the measured error of a photonic interferometer, which it captures when few photons are available, together with the Ziv–Zakai bound [29, 30], which also holds for a prior with hard edges [31]. In simulations of iterative single-molecule localization, where each measurement is placed using the previous estimate, it has been used to compare placement strategies [32]. Such a bound has not yet been applied to the ODMR search as a whole, starting from nothing more than the search window. It would give a strategy-independent reference, showing how close current strategies come to the best achievable performance and which of their measurements are unavoidable. It would also supply a mechanism for the speedup over a sweep, including its expected growth as the line occupies a smaller fraction of the window [15].

Here we derive this limit for the ODMR search and use it to analyze SBED and drive-frequency sweeps for continuous-wave (CW) ODMR. The limit follows from Bayesian estimation theory with the search window as the prior: a Ziv–Zakai bound [29, 30] for finding the resonance, where the window's hard edges leave the van Trees bound empty, and the van Trees bound [27, 28] once it is found, where the information it counts is the same quantity that SBED maximizes. Together they cover the whole run: finding the resonance, narrowing in on it and refining the estimate are parts of one curve rather than separate mechanisms. The same bound gives the speedup over a sweep: for a precise target it is approximately the ratio of the window width to the linewidth, independent of the signal-to-noise ratio, which explains its growth with the window. It also recovers the standard CW-ODMR sensitivity as the refine limit, reached only by a strategy that measures at the steepest slope of the line, and shows that the window limits the bandwidth while the line shape and the signal-to-noise ratio limit the sensitivity.

Section II derives the bound from the physical model, together with the speedup, sensitivity and bandwidth that follow from it. Section III describes SBED, with its particle filter, as a strategy that follows the bound, and the sweeps it is compared with, both evaluated against the bound. Section IV presents the simulations and the experiment, and Sec. V discusses the limits of the approach. The Supplemental Material [33] gives the ODMR

line shapes, the derivations behind the bound for a search window and behind the expected information gain in each regime, and the algorithm settings.

II. THEORY

A. Physical model

Under green illumination the negatively charged N-V center emits red photoluminescence (PL). A center in the ground-state spin sublevels $m_s = \pm 1$ crosses more often into a dark metastable singlet than one in $m_s = 0$, so it emits less PL, and the singlet decays preferentially back to $m_s = 0$ [3, 34]. Continuous illumination therefore pumps the spin into $m_s = 0$, and the spin state can be prepared and read out optically (Sec. S1 of the Supplemental Material [33]).

The ground state is a spin 1. With a static field B_z along the N-V axis, its Hamiltonian is [3]

$$H = D\,S_z^2 + \gamma_e B_z S_z + A_\parallel S_z I_z, \tag{1}$$

written in frequency units ($h = 1$), with $D \approx 2.87$ GHz the zero-field splitting, $\gamma_e \approx 28$ GHz/T the electron gyromagnetic ratio, and $A_\parallel$ the axial hyperfine coupling to the nitrogen nuclear spin I ($|A_\parallel| = 2.15$ MHz for ^{14}N, $I = 1$, and 3.03 MHz for ^{15}N, $I = 1/2$ [35]). The eigenstates are the product states $|m_s, m_I\rangle$, and the magnetic-dipole transitions $m_s = 0 \to \pm 1$, which conserve m_I, lie at

$$f_\pm(m_I) = D \pm \left(\delta_Z + A_\parallel m_I\right), \qquad \delta_Z = \gamma_e B_z, \tag{2}$$

where the Zeeman shift δ_Z is the distance of each transition from D: the field moves the two transitions apart, symmetrically about D. The hyperfine term splits each of them further into $2I+1$ lines, one per nuclear projection m_I, spaced by $|A_\parallel|$: a triplet for ^{14}N and a doublet for ^{15}N. Every N-V center contains one of the two isotopes (^{14}N at 99.6% natural abundance, ^{15}N in isotopically enriched or implanted samples), so the hyperfine structure is always present; whether it is resolved depends on the linewidth.

An ODMR measurement sets the microwave drive frequency f and records the PL, normalized to its off-resonance level. When f meets a transition, the drive transfers population from $m_s = 0$ to the darker $m_s = \pm 1$ state and the PL dips. When the two transitions, $2\delta_Z$ apart, are well separated compared with the linewidth, a drive near f_- couples only $m_s = 0$ and $m_s = -1$, and the spin reduces to a driven two-level system [36]. It is therefore enough to model one transition, and we take the one measured in our setup, f_- (Sec. III D); f_+ behaves the same way, mirrored about D. With the zero-field splitting D treated as a known constant, f_- carries the field, $D - f_- = \delta_Z = \gamma_e B_z$, and an uncertainty in f_- is the same uncertainty in the Zeeman shift.

In continuous-wave (CW) ODMR the laser and the microwave drive act on the spin at the same time. The laser pumps the spin back toward $m_s = 0$ at a rate Γ_p, and each optical cycle destroys the phase between $m_s = 0$ and $m_s = -1$, at a rate Γ_c; both rates grow with the laser power and saturate at high power. The spin's environment adds its own relaxation rate $1/T_1$ to the first and its dephasing rate $1/T_2$ to the second, so the populations relax at $\Gamma_1 = \Gamma_p + 1/T_1$ and the coherence at $\Gamma_2 = \Gamma_c + 1/T_2$. With these rates, and the angular Rabi frequency $\Omega_R = 2\pi\Omega$ of the drive, the steady-state line is a Lorentzian (Sec. S1 D of the Supplemental Material [33]), of half width [36]

$$\gamma = \frac{1}{2\pi}\sqrt{\Gamma_2^2 + \Omega_R^2\,\frac{\Gamma_2}{\Gamma_1}}. \tag{3}$$

At weak drive the laser sets the width, at strong drive the microwave. The spin bath also spreads the transition frequency itself, by $\sim 1/T_2^*$; this adds a Gaussian broadening to the Lorentzian, and it sets the width once both powers are low (Sec. S1 D of the Supplemental Material [33]). Whatever its origin, the line is one dip of a single effective width, taken here as a Lorentzian: the normalized PL expected at drive frequency f is

$$\mu(f) = 1 - C\,\frac{\gamma^2}{\gamma^2 + (f - f_-)^2}, \tag{4}$$

with contrast C, the depth of the dip at $f = f_-$, half width γ of Eq. (3) and full width at half maximum $\Gamma = 2\gamma$. Optical and microwave broadening are not modeled separately: they enter only through the single linewidth Γ, which is inferred with the line.

Equation (4) is a single line. It holds when the hyperfine lines merge, $\Gamma \gg |A_\parallel|$, or when the nuclear spin is polarized into one m_I [37].

Noise. Each reading of the normalized PL is noisy. Its floor is photon shot noise: a reading that collects N_{ph} photons on average off resonance has a relative standard deviation $N_{\mathrm{ph}}^{-1/2}$, and slightly less inside the dip, where fewer photons arrive. Technical noise (laser intensity, microwave power, collection drift) adds to it [38]. We model a single reading, a shot, as

$$y = \mu(f) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{5}$$

with one $\sigma \geq N_{\mathrm{ph}}^{-1/2}$ at every drive frequency; this neglects the small reduction of the shot noise inside the dip, of relative order of the contrast C. A measurement is the mean of k shots at the same f, so its noise is $\sigma/\sqrt{k}$. The signal-to-noise ratio of a single shot depends on where it is taken: it is the drop in PL at f over the noise,

$$\mathrm{SNR}(f) = \frac{1 - \mu(f)}{\sigma} = \frac{C}{\sigma}\,\frac{\gamma^2}{\gamma^2 + (f - f_-)^2}. \tag{6}$$

It is largest on resonance, where the drop is the contrast C,

$$\mathrm{SNR} \equiv \mathrm{SNR}(f_-) = C/\sigma, \tag{7}$$

the contrast over the noise, the single number by which a line is labeled. All SNR values in this paper follow Eq. (7).

The prior knowledge of the experimentalist is an interval of width W that holds the resonance, $f_- \in [f_{\min}, f_{\min}+W]$, and the drive frequency is scanned over at least this interval. The ratio W/Γ, the inverse of the fraction of the window the line occupies, sets how hard the search is.

The model above predicts what a measurement records for given parameters. The rest of this section turns the question around: given the measurements, how well can f_- be known, how much can each measurement add, and where should the next one be taken.

B. Information of a measurement

We use the notation of experiment design [39]: the design $d \equiv f$ is the drive frequency at which a measurement is taken, the outcome y is the normalized PL it records, the parameter of interest $\theta \equiv f_-$ is the transition frequency, and $\phi = (\Gamma, C)$ are the nuisance parameters. The expected normalized PL $\mu(d;\theta,\phi)$ of Eq. (4), now written as a function of the design and the parameters, is the mean outcome, and Eq. (5) gives the spread about it.

A measurement at drive frequency d returns one number y, the normalized PL averaged over its k shots. The model predicts the signal $\mu(d;\theta,\phi)$, and by Eq. (5) y scatters about it with variance σ^2/k, the same whatever θ is. The likelihood $\mathcal{L}(\theta) = p(y \mid d, \theta)$ is the probability of that outcome if the dip is at θ. For the Gaussian scatter of Eq. (5),

$$\mathcal{L}(\theta) = \sqrt{\frac{k}{2\pi\sigma^2}}\, \exp\Big[-\frac{k\,\big(y-\mu(d;\theta,\phi)\big)^2}{2\sigma^2}\Big].$$

How much one measurement at d tells about θ is its Fisher information. With the score $\partial \ln\mathcal{L}/\partial\theta = (k/\sigma^2)(y-\mu)\,\partial\mu/\partial\theta$ it is

$$\begin{aligned} I(d) &= \Big\langle \Big(\frac{\partial \ln\mathcal{L}}{\partial\theta}\Big)^2 \Big\rangle_y = \Big(\frac{k}{\sigma^2}\,\frac{\partial\mu}{\partial\theta}\Big)^2 \underbrace{\big\langle (y-\mu)^2 \big\rangle_y}_{\sigma^2/k} \\ &= \frac{\overbrace{k}^{\text{shots}}}{\underbrace{\sigma^2}_{\text{shot variance}}}\, \overbrace{\Big(\frac{\partial\mu(d;\theta,\phi)}{\partial\theta}\Big)^2}^{\text{slope}^2}, \end{aligned} \tag{8}$$

the average running over the outcomes y the measurement could return, and the slope $\partial\mu/\partial\theta$ being how fast the predicted signal moves when θ moves.

For the Lorentzian of Eq. (4) the slope vanishes on resonance, where the dip is flat, and on the baseline far from it: a measurement at the bottom of the dip or far outside it says almost nothing about where the dip is. The slope is steepest on the two sides of the dip, a detuning $\gamma/\sqrt{3}$ from the resonance, where the dip is 3/4 deep (Appendix A). A measurement there carries the most information one measurement can,

$$I_{\max} = \max_d I(d) = \frac{27}{16}\, k\, \frac{\mathrm{SNR}^2}{\Gamma^2}, \tag{9}$$

with $\mathrm{SNR} = C/\sigma$ of Eq. (7). The best place to measure is thus the flank of the dip, not its center, and the information there grows as $(\mathrm{SNR}/\Gamma)^2$: a deeper or a narrower line is equally valuable. $I_{\max}$ assumes the nuisance parameters (width, contrast) are known; with free nuisances it is an upper bound on the usable information. The nuisances are inferred along with θ but are not the subject of this work.

Measurements are independent given θ, so their Fisher information adds: n measurements at $d_1, \ldots, d_n$ carry

$$J_n = \sum_{i=1}^{n} I(d_i) \;\le\; n\, I_{\max},$$

with equality only when every measurement is taken at the steepest slope of the true line. By the Cramér–Rao bound [40], an unbiased estimate of θ from them has a standard deviation of at least $J_n^{-1/2} \ge (n I_{\max})^{-1/2}$.

C. Prior information

Fisher information measures knowledge in units of an inverse variance of θ, which is Hz^{-2} since θ is a frequency: a Gaussian estimate of standard deviation σ_θ is worth $J = \sigma_\theta^{-2}$. The same currency measures what is known *before* the first measurement, and that depends on the task.

In *tracking* the resonance was measured a short time ago and has since moved by a random amount, so the prior is a Gaussian of standard deviation $\sigma_0^{\mathrm{track}}$ around the last estimate, narrow compared with the window. In a *global search* the resonance is known only to lie somewhere in the window of width W, and the prior is uniform on the window, $p(\theta) = 1/W$.

The information of a prior probability density function (PDF) $p(\theta)$ is its Fisher information. For the Gaussian prior of tracking,

$$J_0^{\mathrm{track}} = \int \frac{p'(\theta)^2}{p(\theta)}\, d\theta = \big(\sigma_0^{\mathrm{track}}\big)^{-2}. \tag{10}$$

It is knowledge the strategy starts with: a narrower prior holds more of it, and no measurement had to pay for it.

D. Van Trees bound

What we compare the strategies against is a limit on the error, whatever its source, the prior or the measurements. The van Trees inequality [27, 28] adds the information of the two, and with $I(d) \le I_{\max}$ gives

$$\bar\sigma_\theta(n) \;\ge\; \sigma_{\mathrm{vT}}(n) = \big(J_0 + n I_{\max}\big)^{-1/2}, \tag{11}$$

for any estimator, biased or not, with $J_0 = J_0^{\text{track}}$ in tracking. Here $\bar{\sigma}_\theta^2(n) = \mathbb{E}\big[(\hat{\theta}_n - \theta)^2\big]$ is the mean squared error (MSE) of the estimate $\hat{\theta}_n$ after n measurements. The bar marks the average: over the noise of the measurements and over the true θ drawn from the prior, here the random drift since the last estimate. It is an error against the true θ, not the uncertainty a strategy claims. For a signal linear in θ the bound is reached exactly; for the dip it holds at every n and is reached once the estimate is within the line.

The curve has two limits, set by how the information of the measurements compares with that of the prior. While $nI_{\max} \ll J_0$, it stays at the prior, $\sigma_{\text{vT}} \approx \sigma_0^{\text{track}}$: the measurements have not yet added to what the prior told. An estimate in this limit leans on the prior and is biased toward its center, which is why its error may lie below the unbiased Cramér–Rao bound $(nI_{\max})^{-1/2}$. Once $nI_{\max} \gg J_0$, the prior is washed out and $\sigma_{\text{vT}} \approx (nI_{\max})^{-1/2}$, the unbiased Cramér–Rao bound of Sec. II B, which falls with slope $-\frac{1}{2}$ on a log–log scale and is reached only by measuring every time at the steepest slope. The two limits cross at $n_c = J_0/I_{\max}$.

Any strategy therefore starts in the prior limit and, if it does well, ends in the data limit, and in between it must pass from one to the other. We name these regimes after what the strategy does in them: *search* while it has yet to find the dip, so that its measurements carry little information and its error stays near the prior; *refine* once it measures on the sides of a dip it has located, so that they carry about $I_{\max}$ and its error follows the data limit; and *localize* in between, from the first detection of the dip until it locks onto the dip, as it narrows down from the window to the dip. A tracking run with $\sigma_0^{\text{track}} \lesssim \Gamma$ has the dip located from the start, so it has no search and is all refine.

E. Global search

In a global search the best guess before any measurement is the center of the window, and its error is the spread of the uniform prior,

$$\bar{\sigma}_\theta^2(0) = \big(\sigma_0^{\text{global}}\big)^2 = \int_{-W/2}^{W/2} \frac{v^2}{W}\, dv = \frac{W^2}{12}, \tag{12}$$

with v the distance from the center of the window. Eq. (11) cannot describe what follows, for two reasons. First, the uniform prior steps from $1/W$ to zero at the edges of the window, where p' diverges, so

$$J_0^{\text{global}} = \int \frac{p'(\theta)^2}{p(\theta)}\, d\theta = \infty,$$

and the bound becomes empty. Second, even with the inverse variance $12/W^2$ in its place, the bound credits every measurement with $I_{\max}$, and its crossover,

$$n_c = \frac{12/W^2}{I_{\max}} = \frac{64}{9}\,\frac{(\Gamma/W)^2}{k\,\text{SNR}^2} \overset{W \gg \Gamma}{\ll} 1,$$

would put the data limit at the first measurement. But $I_{\max}$ is reached only once the dip has been found: a measurement on the baseline carries $I(d) \approx 0$. What such a measurement contributes instead is *elimination*: it rules out the part of the window where a dip would have shown.

Elimination model. The simplest strategy built on elimination steps through the window from one edge, only until it finds the dip, and from then on measures only at the steepest slope. This is not a sweep, which keeps spreading its measurements over the whole window after the dip is found (Sec. II H). Let the SNR be so high that a single measurement shows a dip within its reach Δd, the width around the drive frequency in which the drop cannot be mistaken for noise, and none further away. The steps are spaced by this reach: closer spacing repeats work, wider spacing lets the dip slip between them. Within the line the reach is about Γ; at high SNR the Lorentzian tails stay visible further out, and $\Delta d > \Gamma$ (Sec. S2 of the Supplemental Material [33]). In one run, after n measurements without a dip, the resonance is known only to lie in the part of the window not yet excluded, and the posterior is uniform there, so the error of that run is

$$\sigma_\theta(n) = \frac{W - n\Delta d}{\sqrt{12}}, \qquad n < n_d, \tag{13}$$

where $n_d \le W/\Delta d$ is the step that finds the dip. Detection localizes the resonance to one step, an interval of width Δd and inverse variance $12/\Delta d^2$, after which each measurement adds $I_{\max}$:

$$\sigma_\theta(n) = \Big(\frac{12}{\Delta d^2} + (n - n_d)\, I_{\max}\Big)^{-1/2}, \qquad n \ge n_d. \tag{14}$$

A single run thus has a plateau, a cliff at its own n_d and a $-\frac{1}{2}$ tail (Fig. 1). Where the cliff falls depends only on where the dip happens to lie, possibly under the first measurement.

To average over where the dip lies, put the window at $[0, W]$ with the steps starting at 0. After n steps a dip at $\theta > n\Delta d$ is not yet found, and its run estimates the center of the part not yet excluded, $(W + n\Delta d)/2$; a dip at $\theta < n\Delta d$ has been found, with the error $\sigma_\theta^2(n \mid \theta)$ of Eq. (14). With θ uniform on the window,

$$\begin{aligned}\bar{\sigma}_\theta^2(n) &= \frac{1}{W}\int_{n\Delta d}^{W} \Big(\theta - \frac{W + n\Delta d}{2}\Big)^2 d\theta \\ &\quad + \frac{1}{W}\int_0^{n\Delta d} \sigma_\theta^2(n \mid \theta)\, d\theta \\ &= \frac{(W - n\Delta d)^3}{12\,W} + \frac{1}{W}\int_0^{n\Delta d} \sigma_\theta^2(n \mid \theta)\, d\theta.\end{aligned} \tag{15}$$

The first term holds the runs still searching, a fraction $(W - n\Delta d)/W$, each with the error of Eq. (13); the second, the runs that have found the dip, is small during the search. The mean is dominated by the runs still searching, so it stays near the prior until most of the window is excluded, later than the cliff of a typical run. In inverse

variance, $12W/(W-n\Delta d)^3$, it gains more per step as the part of the window not yet excluded shrinks, not a fixed amount per measurement.

No search is faster. At high SNR the elimination model is not only the simplest search, it is the best one. The Ziv–Zakai bound [29, 30] limits the mean error by how often the data cannot tell a dip at θ from one at $\theta + h$, weighted by h. It needs no smooth prior, and it has been used for this reason with priors that have hard edges [31]. Before any measurement no two positions can be told apart, and the bound gives exactly Eq. (12). For a dip that a measurement detects only within reach of it, every outcome before the first detection looks like the baseline, so any strategy, adaptive or not, is up to that point a fixed order of probes, and two positions are told apart once either of them has been probed. The bound is then smallest when the probed part of the window is contiguous, and equals the first term of Eq. (15) (Sec. S2 of the Supplemental Material [33]).

Search at lower SNR. At lower SNR one measurement in the dip no longer reveals it, and the elimination model needs repeats: a probe position is excluded only once enough shots there look like the baseline. Each step is then a detection, with a miss and a false alarm at every position. Averaging the same number of shots at every position keeps false alarms rare over all $W/\Delta d$ positions only by raising the threshold with their number, which costs about $\ln(W/\Delta d)$ times more shots. A sequential test spends its shots where they are needed: it drops a position as soon as it looks like the baseline and measures again only where the outcome is ambiguous. It pays the price of a rare false alarm once, at the position of the dip, and for a large window it removes the logarithm [41]. A Bayesian strategy goes further: its posterior combines the outcomes at neighboring positions, which the tails of the dip reach as well, and lowers the weight of a position with each outcome that looks like the baseline instead of excluding it at once.

No strategy can search faster than a bound of the same kind as Ziv–Zakai, which compares a dip at θ with a world that has no dip at all. How well one measurement tells the two apart is the Kullback–Leibler divergence between its outcomes in the two worlds (Sec. II G), $k\, s(d;\theta)^2/2\sigma^2$ for a dip of depth $s(d;\theta)$ at the drive frequency: the squared SNR of the drop it would see. In the world without a dip the strategy's choices cannot depend on θ, so, averaged over where the dip could be, a measurement tells them apart by at most

$$\bar{\kappa} = \frac{1}{W}\int \frac{k\, s(d;\theta)^2}{2\sigma^2}\, d\theta = \frac{\pi\gamma\, k\,\mathrm{SNR}^2}{4W}.$$

The fraction of runs that have not yet located the dip therefore falls no faster than about $e^{-n\bar{\kappa}}$ [42]. Together with the data limit, this bounds the mean error of any strategy, adaptive or not, at any SNR and every n (Sec. S2 of the Supplemental Material [33]):

$$\bar{\sigma}_\theta^2(n) \;\geq\; \max\left\{ \frac{W^2}{8e}\,\frac{e^{-n\bar{\kappa}}}{(1+\ln 2+n\bar{\kappa})^2}\,,\; \max_{m\geq 2}\frac{1-1/m}{\pi^2 m^2/W^2 + nI_{\max}} \right\}, \tag{16}$$

with m an integer. The first term is the search. It stays of the order of the prior until $n\bar{\kappa}$ is of order one, so any strategy needs at least about

$$n_{\mathrm{search}} \;\gtrsim\; \frac{1}{\bar{\kappa}} = \frac{8}{\pi}\,\frac{W}{\Gamma}\,\frac{1}{k\,\mathrm{SNR}^2} \tag{17}$$

measurements to find the dip. The second term is the data limit of a uniform window, which tends to $(nI_{\max})^{-1}$ once $nI_{\max}W^2 \gg 1$. The sequential test above finds the dip after a number of measurements of the order of Eq. (17), so at low SNR the bound is tight up to a constant. At high SNR Eq. (17) allows less than one measurement per line width, which no strategy can take: with $\Delta d \approx \Gamma$, $1/\bar{\kappa}$ exceeds $W/\Delta d$ only for $k\,\mathrm{SNR}^2 < 8/\pi$, an SNR of about 1.6 at $k = 1$. Above it the first term of Eq. (15), exact for a dip of finite reach, is the tighter bound, and the elimination model meets it. Before the dip is found, no strategy knows where to look, and none gains more per measurement than the window average.

After detection. Once the dip is found, the posterior is a single smooth peak around it, shaped by the measurements that saw the dip, and the problem has become tracking (Sec. II C) with that posterior as the prior. Van Trees applies again from there: Eq. (14) is Eq. (11) restarted at detection, with J_0 the information of the posterior, $12/\Delta d^2$ in the model (Sec. S2 of the Supplemental Material [33]). It covers localize and refine alike. In localize a strategy is not yet measuring at the steepest slope, its posterior may still have several peaks, and its error lies above the curve; refine starts when its error runs parallel to it. In the elimination model localize is the single detection step. A dip at θ is found at step $n_d = \lceil \theta/\Delta d \rceil$, so inserting Eq. (14) into the second term of Eq. (15), one step of width Δd per n_d, gives the mean error of the model at every $n \leq W/\Delta d$,

$$\bar{\sigma}_\theta^2(n) = \frac{(W-n\Delta d)^3}{12\,W} + \frac{\Delta d}{W}\sum_{n_d=1}^{n}\left(\frac{12}{\Delta d^2} + (n-n_d)\, I_{\max}\right)^{-1}, \tag{18}$$

with the sum ending at $W/\Delta d$ for larger n. It starts at Eq. (12) and tends to the data limit $(nI_{\max})^{-1}$ at large n. In a window about as narrow as the line the first measurements already land in the dip, so there is no plateau (Fig. 1, dashed curve). A sweep has none of the three regimes, since it gains the same information I_{sweep} at every step (Sec. II H).

Reading the bounds. We judge a strategy by how close its error comes to these curves. Every bound here limits a mean of $(\hat{\theta}_n - \theta)^2$, and they differ in what varies inside that mean, which decides how a strategy is read against each. The unbiased Cramér–Rao bound averages over the noise at a fixed θ.

Van Trees in tracking averages also over θ, but only over the drift since the last estimate. The error averaged over runs is compared with Eq. (11); a gap means that the measurements carried less than $I_{\max}$, or that the strategy did not use them efficiently.

The search bound averages over θ anywhere in the window: over runs whose dips lie at different places. It therefore constrains only the mean over runs, and the gap of that mean above it measures how fast the strategy covers the window. A single run says nothing against it, since it stays on its plateau and drops at its own n_d, wherever its dip happens to be.

The restarted van Trees, once a run has found its dip, averages over the noise and over θ only within that run's own narrow posterior, so θ is nearly fixed. It applies to each run, given what the run has measured so far. The slope of a run's error then gives the information each of its measurements carries, a fraction $q \leq 1$ of $I_{\max}$ (Sec. II H), and a gap that does not close means $q < 1$. The restart must be decided from the run's own data, never from the true θ.

For the posterior mean the MSE equals the expected posterior variance, whatever the shape of the posterior. The posterior standard deviation can therefore stand in for the error, averaged over runs during the search and run by run after detection, as long as the posterior is calibrated. A claimed uncertainty below the restarted curve by more than the scatter of the runs points to an overconfident posterior; a single run's posterior variance may fluctuate slightly below it, since the bound is on an expected error.

The bounds treat the width Γ and the contrast C as known. Inferring them as well can only add to the uncertainty in θ, but it adds nothing when the measurements are placed symmetrically on the two sides of the dip: a shift of θ then raises the signal on one side and lowers it on the other, while a change of Γ or C changes both sides alike, so the data tell the two apart.

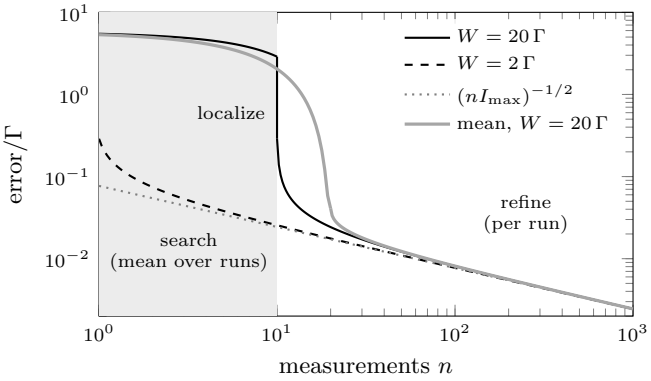


FIG. 1. Schematic of the three regimes of an adaptive strategy, drawn with the elimination model, Eqs. (13)–(18); a Bayesian adaptive strategy follows the same profile, with a less sharp localize step. Illustrative values (SNR 10, $k = 1$, reach $\Delta d = \Gamma$). Black: single runs, solid for a wide window, $W = 20\,\Gamma$, with the dip found halfway through, dashed for a narrow one, $W = 2\,\Gamma$, found at the first step; dotted: unbiased Cramér–Rao bound $(nI_{\max})^{-1/2}$. The wide-window run shows a search plateau (shaded), a localize cliff at detection (vertical segment) and the $-\frac{1}{2}$ refine tail; the narrow window shows no plateau. Gray: the root of the mean squared error over runs of the wide window, Eq. (18), which stays near the prior until most of the window is covered, because the runs that have not yet found the dip dominate it; each run drops at its own n_d. During the search only this mean is bounded, while after detection each run is read against its own restarted van Trees curve, the refine tail. All curves start near the prior width $\sigma_0^{\text{global}} = W/\sqrt{12}$ and tend to the data limit at large n. Simulated and measured runs are shown against the bounds in Sec. IV.

F. Bayesian adaptive strategy

A Bayesian strategy such as SBED is defined quite differently from the elimination model: it keeps a posterior over θ and picks each drive frequency by the expected information gain (Sec. II G), with no grid and no detection rule. Its error nevertheless follows the same three-part profile. Before the dip is found, a measurement on the baseline only removes the posterior mass near where it was taken, so the posterior shrinks by elimination, as in the model; once a measurement lands in the dip the posterior collapses onto it; and once the dip is located, the strategy measures at the steepest slope and reaches the data limit (Sec. II G). The model is therefore a picture of a Bayesian run, not a prediction of its step counts: at low SNR one measurement no longer reveals the dip, n_d grows, and localize spreads over many measurements. The runs are read against the bounds rather than the model: during the search through their mean, Eqs. (15)–(17), which hold for any strategy, and after each run's own detection against Eq. (11) restarted there.

G. Information gained per measurement

The bound counts information as Fisher information, an inverse variance of θ. A Bayesian strategy chooses its measurements by a different quantity, how much a measurement is expected to change the posterior. This section defines that quantity, the form of it that SBED computes, and when maximizing it reaches the bound.

Expected information gain. How much a distribution q differs from a distribution p is measured by the Kullback–Leibler divergence [43], $D_{\mathrm{KL}}(q \,\|\, p) = \int q \ln(q/p)$, which is zero only when the two agree. What a measurement at d contributes is the divergence of the posterior after it from the posterior before it. Before the outcome is known, its expectation over the outcomes y the measurement could return is the expected informa-

tion gain [44],

$$\begin{aligned}\mathrm{EIG}(d) &= \mathbb{E}_{y|d}\Big[D_{\mathrm{KL}}\big(p(\theta,\phi \mid y,d)\,\|\,p(\theta,\phi)\big)\Big] \\ &= I(\theta,\phi;y \mid d) \\ &= \underbrace{H[y \mid d]}_{\text{predictive entropy}} - \underbrace{\mathbb{E}_{\theta,\phi}H[y \mid \theta,\phi,d]}_{\frac{1}{2}\ln(2\pi e\,\sigma^2/k)\ \text{at every } d}\ , \end{aligned} \tag{19}$$

the mutual information [45] between the parameters and the outcome, with $p(\theta,\phi)$ the current posterior and H the entropy. The second term is the entropy of the noise of Eq. (5), so all the dependence on d is in the predictive entropy.

The variance form computed by SBED. The predictive distribution of y has two spreads, the noise σ^2/k and the spread of the predicted signal over the posterior, $\sigma^2_{\mathrm{pred}}(d) = \mathrm{Var}_{\theta,\phi}[\mu(d;\theta,\phi)]$. Its exact entropy needs the whole distribution. SBED replaces it with a Gaussian of the same variance, which has the largest entropy for a given variance [45], so the result is an upper bound on the gain,

$$\mathrm{EIG}(d) \le \tfrac{1}{2}\ln\Big(1+\frac{k\,\sigma^2_{\mathrm{pred}}(d)}{\sigma^2}\Big), \tag{20}$$

exact when the predicted signal is Gaussian distributed over the posterior. SBED (Sec. III B) evaluates Eq. (20) at every candidate drive frequency and measures where it is largest, the greedy rule

$$d_{n+1} = \arg\max_d \mathrm{EIG}_n(d), \tag{21}$$

with EIG_n evaluated on the posterior after n measurements.

Two questions. First, when is the Gaussian form of Eq. (20) equal to the exact gain? Second, the van Trees bound of Eq. (11) is reached only by measuring at the steepest slope, yet the greedy rule of Eq. (21) maximizes the expected gain and never refers to the slope: when does it put the measurements there? In refine both hold, asymptotically in n; in search and localize neither does.

Refine: the gain becomes Fisher information. Once the dip is located, the posterior of θ is a single peak, narrow compared with the line, so a first-order Taylor expansion about the estimate $\hat\theta$ makes the signal linear in θ: $\mu(d;\theta) \approx \mu(d;\hat\theta) + (\partial\mu/\partial\theta)(\theta-\hat\theta)$. Then $\sigma_{\mathrm{pred}}(d) = |\partial\mu/\partial\theta|\,\sigma_\theta$ whatever the shape of the peak, and the rule measures where the Fisher information $I(d)$ of Eq. (8) is largest, at the steepest slope. With enough data the peak also becomes Gaussian by the central limit theorem for posteriors (the Bernstein–von Mises theorem [46]), of variance $\sigma^2_\theta \approx 1/J_n$, so Eq. (20) is an equality and

$$\begin{aligned}\mathrm{EIG}(d) &= \tfrac{1}{2}\ln\Big(1+\underbrace{\frac{k}{\sigma^2}\Big(\frac{\partial\mu}{\partial\theta}\Big)^2}_{I(d)}\underbrace{\sigma^2_\theta}_{1/J_n}\Big) \\ &= \tfrac{1}{2}\ln\frac{\overbrace{J_n + I(d)}^{J_{n+1}}}{J_n} = \ln\frac{\sigma_{\theta,n}}{\sigma_{\theta,n+1}}\ : \end{aligned} \tag{22}$$

the measurement adds $I(d)$ to the information, and the EIG is the logarithm of the shrinkage of the posterior width. The variance form and the exact gain then pick the same point: the Gaussian form is exact here. Each gain is a mutual information, Eq. (19), so by the chain rule of mutual information [45] the expected gains of a run add up to the information all its outcomes carry together. In refine each depends on its measurement only through $I(d_i)$, so maximizing every $I(d_i)$ also maximizes the total: greedy EIG is optimal over the whole run and reaches the data limit of Eq. (11): greedy choice does put the measurements at the steepest slope (Sec. S3 of the Supplemental Material [33]). Both results need the posterior to be one peak with $\sigma_\theta \ll \gamma$ and almost no mass left elsewhere in the window, which holds in refine and not at the start of localize (Sec. S4 of the Supplemental Material [33]).

Search and localize. Before the dip is located the posterior of θ is spread over the window, the predicted signal at d is two-valued (the dip is there or not), and neither result holds (Sec. S3 of the Supplemental Material [33]). The Gaussian form is no longer exact: it ranks search candidates as the exact gain does but overvalues them against refine candidates, and in localize it keeps spending measurements on residual posterior mass far from the dip, which lowers the MSE rather than the entropy. Nor is greedy choice optimal over the run: where one measures now changes what later measurements can gain, so choosing the most informative next measurement need not give the most informative run. The bound is therefore met in refine, and the strategies of Sec. III B differ in how they pass search and localize.

H. Speedup over a drive-frequency sweep

The same budget predicts the speedup over a conventional sweep. An adaptive strategy in the refine regime measures at the steepest slope and gains, per measurement, Eq. (9),

$$I_{\max} = \frac{27}{16}\,\frac{k\,C^2}{\sigma^2\,\Gamma^2}.$$

A sweep spreads its measurements uniformly over the window, so the information it gains per step is the window average of Eq. (8), taken over the drive frequency in the window (not over the prior),

$$\begin{aligned} I_{\mathrm{sweep}} &= \frac{1}{W}\int I(f)\,df = \frac{k}{\sigma^2 W}\int\Big(\frac{\partial\mu}{\partial\theta}\Big)^2 df \\ &\overset{W\gg\Gamma}{=} \frac{\pi}{2}\,\frac{k\,C^2}{\sigma^2\,\Gamma\,W}. \end{aligned} \tag{23}$$

Comparing the two rates,

$$I_{\max} \propto \frac{k\,C^2}{\sigma^2}\,\frac{1}{\Gamma^2}, \qquad I_{\mathrm{sweep}} \propto \frac{k\,C^2}{\sigma^2}\,\frac{1}{\Gamma}\,\frac{1}{W}\ :$$

they share the factor kC^2/σ^2 and differ only in the width factor. The extra factor Γ/W of the sweep is the fraction of its measurements that land on the line at all.

Where a sweep measures does not depend on what it has learned, so its information grows at the constant rate I_{sweep}, while an adaptive strategy in refine gains $I_{\max}$ per measurement. For a precise target, where the search cost no longer matters, the speedup is the ratio of the two rates,

$$R_\infty = \frac{I_{\max}}{I_{\text{sweep}}} = \frac{27}{8\pi}\,\frac{W}{\Gamma} \approx 1.07\,\frac{W}{\Gamma}\,: \tag{24}$$

the speedup is the inverse occupancy of the window, the number of linewidths that fit in it, which Dushenko *et al.* [15] identified qualitatively; Eq. (24) gives its prefactor. It is independent of the SNR, because both strategies gain information $\propto \text{SNR}^2$. For a loose target the search cost dominates and the speedup is smaller (Sec. S5 of the Supplemental Material [33]). A sweep cannot approach the van Trees bound at any n: its rate I_{sweep} is fixed by the window, while the bound's data limit requires every measurement at the steepest slope.

Comparing strategies. A strategy is thus described by two numbers: what it pays to find the dip, and the rate at which it gains information afterwards. Both are read off the error averaged over runs, without a detection model; a single run's values depend on where its dip happened to lie and prove nothing on their own. Late in a run the inverse of the averaged squared error grows linearly,

$$\sigma_\theta(n)^{-2} \approx q\, I_{\max}(n - n_0), \tag{25}$$

where $q \le 1$ is the fraction of $I_{\max}$ that the measurements carry, and n_0 is the number of measurements lost, on average, before this growth began. The offset n_0 generalizes n_d, the step that finds the dip in the elimination model of Eqs. (13)–(14): it holds the search and also whatever localizing costs beyond the single step of the elimination model. A sweep has $q = I_{\text{sweep}}/I_{\max} = 1/R_\infty$ and $n_0 \approx 0$, and the ideal adaptive strategy has $q = 1$ and $n_0 \approx n_d$, so all strategies can be placed on one (n_0, q) plane. The number of measurements to a precise target is then $n_0 + \sigma_{\text{tgt}}^{-2}/(qI_{\max})$. In the inverse squared error each strategy is a straight line that leaves the axis at n_0 with slope q (Fig. 2): the sweep starts at once but climbs slowly, an adaptive strategy starts late but climbs fast, so it wins for a precise target and can lose for a loose one.

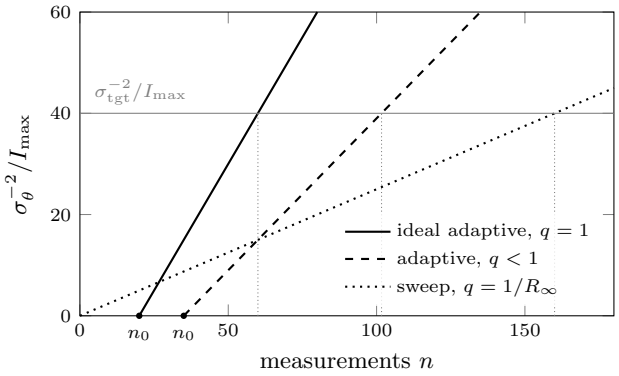


FIG. 2. Schematic of Eq. (25), with exaggerated values chosen for legibility (a real sweep has $q = 1/R_\infty \approx \Gamma/W$, far smaller). In the inverse squared error, in units of $I_{\max}$, each strategy is a line that leaves the axis at its offset n_0 with slope q. Solid: ideal adaptive strategy, $q = 1$, $n_0 = n_d$; dashed: an adaptive strategy that loses information, $q < 1$, larger n_0; dotted: sweep, $n_0 \approx 0$, $q = 1/R_\infty$. The vertical lines mark the measurements each needs to reach a precise target (horizontal line), $n_0 + \sigma_{\text{tgt}}^{-2}/(qI_{\max})$. For a loose target, below the crossings, the sweep is ahead.

I. Sensitivity and bandwidth

Sensitivity and bandwidth follow from the same budget once measurements are converted to time. Let one measurement take $t_m = k\,t_s$ (k shots of duration t_s), so that n measurements take $T = nt_m$. With D known, $D - f_- = \delta_Z = \gamma_e B_z$, so the field uncertainty is $\sigma_B = \sigma_{f_-}/\gamma_e$.

Sensitivity (refine). In the refine regime every measurement adds $I_{\max}$, so after a time T

$$\sigma_{f_-}(T) = (nI_{\max})^{-1/2} = \Big(\frac{T}{t_m}\,I_{\max}\Big)^{-1/2},$$
$$\sigma_{f_-}\sqrt{T} = \sqrt{\frac{t_m}{I_{\max}}} = \gamma_e\,\eta.$$

The uncertainty falls as $\sigma_{f_-} \propto T^{-1/2}$, so the product $\sigma_{f_-}\sqrt{T}$ does not depend on T and characterizes the strategy rather than the length of the run. Converted to field, this constant is the sensitivity η,

$$\begin{aligned}\eta = \sigma_B\sqrt{T} &= \frac{1}{\gamma_e}\sqrt{\frac{t_m}{I_{\max}}} = \frac{1}{\gamma_e}\sqrt{\frac{16\,k\,t_s\,\Gamma^2}{27\,k\,\text{SNR}^2}} \\ &= \frac{4}{3\sqrt{3}}\,\frac{\Gamma\sqrt{t_s}}{\gamma_e\,\text{SNR}},\end{aligned} \tag{26}$$

with $\text{SNR} = C/\sigma$ the single-shot SNR of Eq. (7). The number of shots k cancels as well: averaging k shots into one measurement costs k times the time and gives k times the information. This is the standard CW-ODMR sensitivity of a Lorentzian line [36], here the refine limit of the budget, reached only by a strategy that measures at the steepest slope. A strategy whose refine measurements are not all at the steepest slope gains less per measurement: if they carry on average a fraction $q \le 1$ of $I_{\max}$, $I_{\max}$ is replaced by $qI_{\max}$ in Eq. (26), and its sensitivity is worse by $1/\sqrt{q}$, i.e. $\eta/\sqrt{q}$.

Sensitivity over a run. Read off a run of length T, the product $\sigma_B(T)\sqrt{T}$ is not yet this constant. In tracking, at the bound, $\sigma_{f_-}^{-2} = J_0 + (T/t_m)\,I_{\max}$, so

$$\sigma_B\sqrt{T} = \frac{1}{\gamma_e}\,\frac{\sqrt{T}}{\sqrt{J_0 + (T/t_m)\,I_{\max}}} \longrightarrow \eta,$$

which grows as $\sqrt{T}$ while the prior information J_0 dominates and tends to η once the data dominate. A real strategy reaches the same constant; its search offset n_0 only changes how (Sec. S5 of the Supplemental Material [33]).

Bandwidth (whole run). A field reading is useful once its error reaches the target σ_{tgt}; the time needed, $T(\sigma_{\mathrm{tgt}}) = t_m\,[n_0 + \sigma_{\mathrm{tgt}}^{-2}/(qI_{\max})]$ by Eq. (25), sets the bandwidth $1/T(\sigma_{\mathrm{tgt}})$. For a tight target the refine term dominates and gives the sensitivity of Eq. (26) again. For a loose target the search term dominates, $T \approx n_0 t_m$: the bandwidth cannot exceed about $1/(n_0 t_m)$, whatever the SNR (Sec. S5 of the Supplemental Material [33]). The window, i.e. the prior, therefore limits the bandwidth, while the line shape and SNR limit the sensitivity. Only the total integration time $T = nk\,t_s$ enters the bound, not how it is divided into measurements: n measurements of k shots carry the same information as nk measurements of one shot at the same drive frequency. Neglecting the overhead of switching the drive frequency, the cost of a run is therefore its total number of shots nk, the integration time in units of t_s, and strategies that group their shots differently are compared on that axis.

III. METHODS

A. Bayesian inference

Every strategy runs the same loop, Fig. 3: it chooses a drive frequency d_n, takes one measurement y_n there, updates the posterior over the line parameters with it, and stops when the stop rule of Sec. III B fires. The strategies differ only in the first step; the posterior and its update, described here, are shared. The unknowns are those of Sec. II B, the parameter of interest θ and the nuisances ϕ, together with the noise level σ.

The posterior over (θ, ϕ) is a weighted cloud of particles $\{(\theta_i, \phi_i), w_i\}_{i=1}^{N_p}$, updated by sequential Monte Carlo [47]. Each particle is one hypothesis for the line. It holds the Zeeman shift $\delta_{Z,i}$, so $\theta_i = D - \delta_{Z,i}$ with D fixed (the simulations measure the mirror transition $D+\delta_{Z,i}$, which is equivalent by symmetry), and the nuisances: the width of the line and its contrast C. The mechanisms that set the width, Eq. (3), are not inferred; only the width they produce is. The initial cloud is drawn from the prior.

The noise is Gaussian, Eq. (5), but its level σ is not known in advance, since technical noise adds to the shot noise by an amount that depends on the setup. Each particle therefore carries a closed-form posterior on σ^2, given its line, and σ^2 is integrated out [48, 49]: a measurement reweights each particle by a Student-t likelihood instead of a Gaussian with a plug-in σ, which would drive the estimate of σ to its lower bound. When the weight has concentrated on a few particles, the cloud is resampled [50] and jittered with the Liu–West kernel [51], which preserves its mean and covariance. The noise model, the resampling safeguards and all settings are in Sec. S6 of the Supplemental Material [33].

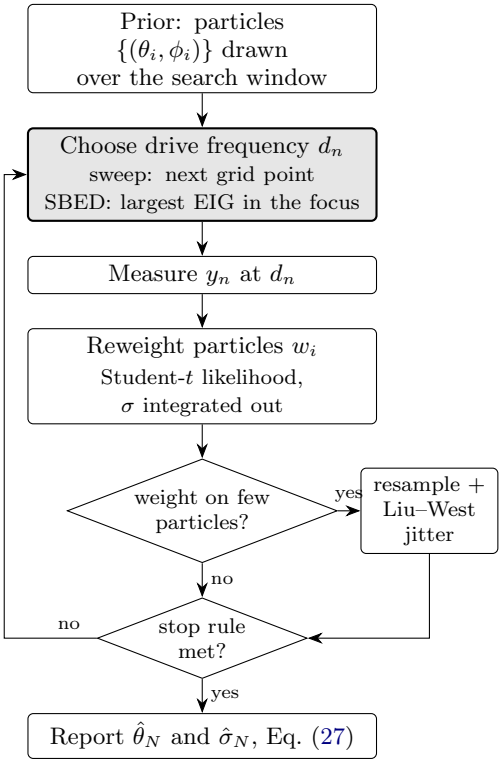


FIG. 3. The measurement loop shared by every strategy. Only the choice of the next drive frequency (shaded) differs between strategies; the particle posterior, its update and the stop rule are the same for all of them. The noise level σ is integrated out in each particle, so the update weighs by a Student-t likelihood.

The particle filter reports the posterior mean and the weighted particle standard deviation,

$$\hat{\theta}_N = \sum_i w_i \theta_i, \qquad \hat{\sigma}_N = \Big[\sum_i w_i (\theta_i - \hat{\theta}_N)^2\Big]^{1/2}, \quad (27)$$

the claimed uncertainty whose meaning Sec. III C discusses.

B. Measurement strategies

We compare Bayesian SBED with the strategy an experimentalist would use without adaptive design, a sweep, run at several grid spacings; the best spacing is the yardstick SBED is measured against. All strategies infer with the particle filter of Sec. III A, so the comparison isolates the choice of measurements from the inference, and $\hat{\sigma}_N$ means the same for all of them. All run under the same stop rule and the same measurement budget: a run stops once the claimed uncertainty of every parameter has stayed below its ceiling for several consecutive measurements and the dip test below finds a dip.

Dip test. The test asks where the measurements, not the particles, show a dip. Each measurement is scored against a baseline taken from the upper part of the data, in units of the noise from the noise posterior. Contiguous stretches of measured drive frequencies, about as wide as a plausible line, get a boxcar score, and a stretch is significant when its score passes a family-wise false-alarm threshold over all stretches tested. Each connected run of significant stretches is one dip.

Sweep. SimpleSweep measures a uniform grid over the window, a few points per FWHM of the prior-mean line, and repeats whole passes; the spacing is the one setting a sweep has, and several are run.

Bayesian SBED. SBED is the particle filter choosing each d by Eq. (20), maximized over candidate drive frequencies concentrated on the steep sides of the posterior-mean dip and on the dips found. The candidates are restricted to the focus: one section per detected dip, padded by one half width on each side, which widens at once when a new dip appears and narrows only after a delay. A strategy that has piled its measurements onto a suspected dip cannot be contradicted elsewhere, so with a small probability, largest while much of the window is unmeasured, SBED instead measures the middle of the widest unmeasured gap, a falsification probe.

C. Simulation setup and evaluation

What the claimed uncertainty is. The particle filter reports $\hat{\sigma}_N$, Eq. (27): a moment of the posterior, defined at every N. During the search its average over runs is compared with the search bound, and after a run's own detection the run is compared with van Trees restarted there (Sec. II E). What fails before the posterior is Gaussian (Sec. II G) is the usual reading of $\hat{\sigma}_N$: with a uniform window or several live candidates it is not a 68% interval, and the posterior entropy is below $\frac{1}{2}\ln(2\pi e\,\hat{\sigma}_N^2)$, so information gains, per-step gains as variance ratios, and coverage at Gaussian quantiles are only meaningful in the data regime. We therefore (i) compare $\hat{\sigma}_N$ with the search bound only as an average over runs, (ii) test coverage, the true error against $\hat{\sigma}_N$, only once the posterior is Gaussian, or with quantiles of the particle distribution itself, and (iii) keep this separate from miscalibration of the particle filter, which makes $\hat{\sigma}_N$ too small at any N.

Use cases. A use case is one simulated signal at one noise level, set by the SNR, the drive window, the Gaussian share of the width and the isotope (Sec. S6 of the Supplemental Material [33]). Every metric is computed within one use case, over repeats whose true signal is the same for every strategy, so strategies are compared run by run; several use cases are summarized by the median over them.

Metrics. The accuracy at the end of a run is not a quality metric: every run stops once its claimed uncertainty is below the ceiling, so its final error mostly reflects the ceiling. We report instead whether the claimed uncertainty is honest, through its coverage, the share of runs whose true error lies within $\hat{\sigma}_N$ and within $2\hat{\sigma}_N$; how close each strategy comes to the bounds of Sec. II E, through the ratio of the search bound, the first term of Eq. (15), to the mean squared true error over runs during the search, and through each run's gap to its restarted van Trees curve after its detection; and the cost, the number of measurements after which the true error stays below a target. After its detection each run is also read through its information rate, the growth of $\hat{\sigma}_N^{-2}$ per measurement in units of $I_{\max}$, Eq. (9); a rate above 1 means the claimed uncertainty fell faster than the data allow. Measurements to stop are compared only between strategies whose claimed uncertainty is equally honest.

D. Experimental setup

We measure the lower transition f_- only: in our setup it needs less microwave power than f_+, which keeps the heating of the stage low and the amplifier within its linear range. [Sample, optics, microwave delivery, detection and the per-shot integration time t_s.]

IV. RESULTS

A. Simulations

[(a) Claimed σ and true error for a narrow and a wide window: the mean over runs against the search bound and Eq. (18), and single runs after detection against restarted van Trees, Eq. (14). (b) Information rate and prior share. (c) Cost vs SNR and window. (d) Calibration of the claimed σ.] [When the SBED posterior becomes Gaussian, and its calibration in refine (Sec. II G).] [Test: share of localize-stage measurements placed more than a few FWHM from the peak, compared with the residual mass there (Sec. II G).]

B. Experiment

[Placeholder until the lab data are in.]

V. DISCUSSION

[Limits: the claimed σ is overconfident; the simulated prior is miscalibrated; Gaussian noise only; $I_{\max}$ assumes known nuisances; low-SNR detection model; the coverage bound is strict only for a dip of finite reach, and at low SNR only the change-of-measure bound holds, about $\ln(W/\Delta d)$ below fixed averaging.]

VI. CONCLUSION

[Conclusion.]

Appendix A: Steepest slope of the Lorentzian line

This appendix derives the best information per measurement, Eq. (9), for the Lorentzian line of Eq. (4). Written in terms of the detuning $u = d - \theta$ of the measurement from the resonance, the line depends on d and θ only through u, so a shift of the resonance acts like the opposite shift of the measurement, and the slope of Eq. (8) is

$$\frac{\partial\mu}{\partial\theta} = -\frac{\partial\mu}{\partial u} = -\frac{2C\gamma^2 u}{(\gamma^2+u^2)^2}.$$

It vanishes at $u = 0$ and as $u \to \pm\infty$. Setting $\partial^2\mu/\partial u^2 = 0$ gives $(\gamma^2+u^2)^2 = 4u^2(\gamma^2+u^2)$, i.e. $u^2 = \gamma^2/3$: the slope is steepest at $|u| = \gamma/\sqrt{3}$, where $\gamma^2/(\gamma^2+u^2) = 3/4$ and the dip is 3/4 deep. There

$$\max_d \left|\frac{\partial\mu}{\partial\theta}\right| = \frac{2C\gamma^2\,(\gamma/\sqrt{3})}{(4\gamma^2/3)^2} = \frac{9C}{8\sqrt{3}\,\gamma} = \frac{9C}{4\sqrt{3}\,\Gamma} \approx 1.30\,\frac{C}{\Gamma},$$

with $\Gamma = 2\gamma$. Inserting this slope into Eq. (8), with $\mathrm{SNR} = C/\sigma$,

$$I_{\max} = \frac{k}{\sigma^2}\Big(\frac{9C}{4\sqrt{3}\,\Gamma}\Big)^2 = \frac{81}{48}\,\frac{k\,C^2}{\sigma^2\Gamma^2} = \frac{27}{16}\,k\,\frac{\mathrm{SNR}^2}{\Gamma^2}.$$

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